

Another Bengaluru based startup, Agdhi, uses a combination of technologies ranging from IoT to machine language and blockchains to detect defective seeds and infect- ed plants. Chennai based Aquacon-

nect is an AI based, full-stack aqua- culture platform that provides farm inputs, market linkage, and financial solutions for shrimp farmers.

Farmonaut and Orbit Field Scout are satellite based crop health moni-

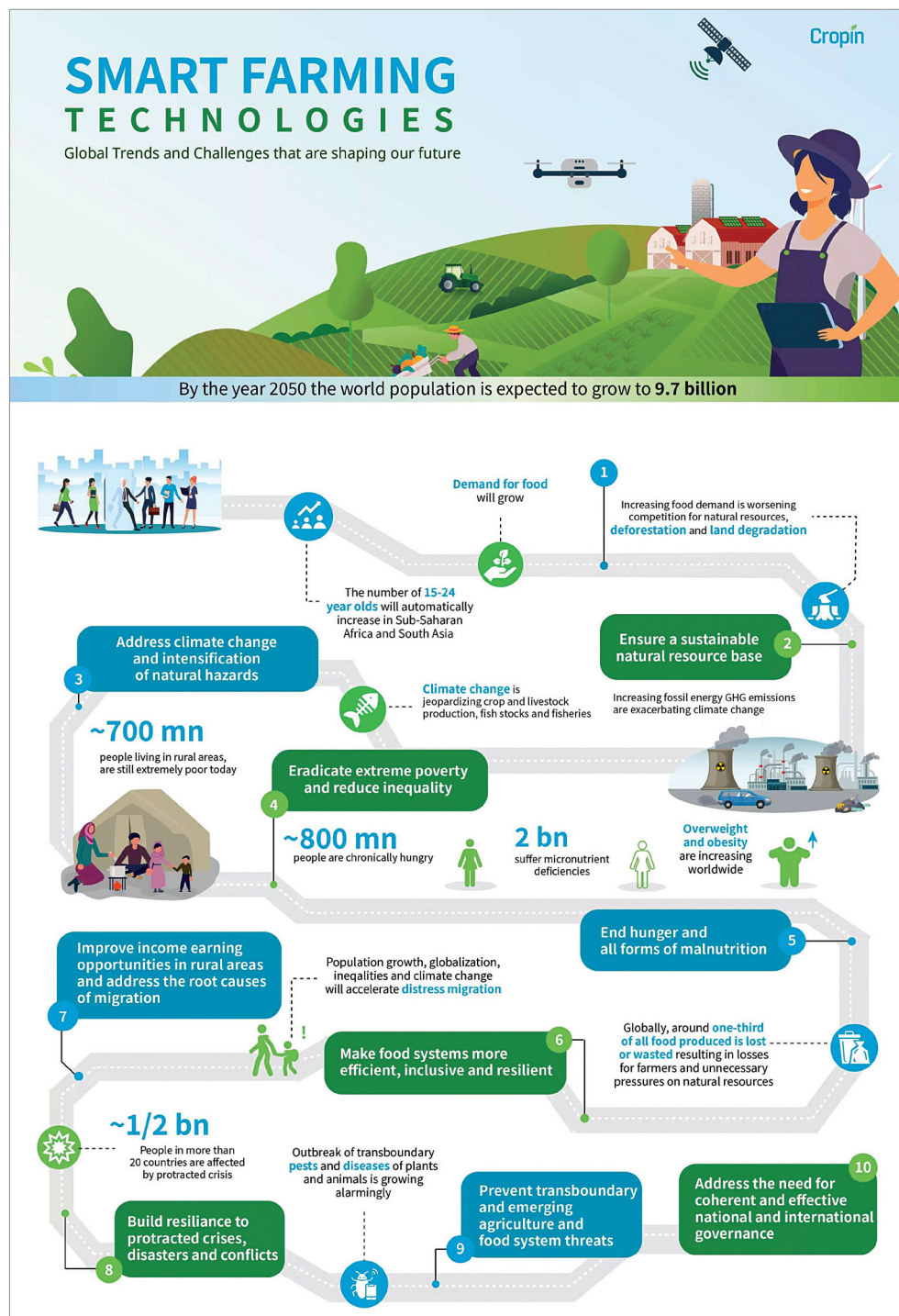
toring solutions, which warn farm- ers early about disease outbreaks in crops, pest attacks by locusts, or animal interference. Dhenutrack is an AI based cow management and tracking system incubated at IIT

Patna. Fyllo and AgricxLab are other AI based portals to watch.

Last year, Micro- soft revealed Project FarmVibes, a new suite of open-sourced, farm- focused technologies from Microsoft Research. The first to be released was FarmVibes.AI, a set of algorithms that run on Mi- crosoft Azure, which can predict the ideal amounts of fertiliser and herbicide a farmer should use and where to apply them; forecast temperatures and wind speeds across fields; inform the farmer when and where he should plant and spray; determine the ideal depth to plant seeds based on soil moisture; and tell how different crops and practices can keep carbon sequestered in the soil. Developers can also enhance these algorithms to suit specific farm needs.

Full-stack. Cropin is a well-known full-stack SaaS solution that helps increase the efficiency, productivity, and sustainability of farms using a combination of technologies ranging from remote sensing to AI, ML, and big data analysis. In September 2022, Cropin announced the launch of Cropin Cloud, a purpose- built industry cloud for agriculture.

Italian confection- ary company Loacker



Smart farming technologies—trends and challenges (Courtesy: Cropin)